

Thank you for your interest in a role with GIC!

We are excited to invite you to the first step of our hiring process: the technical assessment.

Please read the instructions below carefully before starting your solution.

Instructions:

You will implement a falling bricks game. In this game, you will control “bricks” made up of three blocks, each marked with a special symbol. Bricks can be horizontal or vertical, and you can move or drop them using simple commands. The goal is to place all bricks and clear as many matches as possible.

- For your solution, we request that you create a **console application**. Please use the coding language(s) **stipulated in the initial email** for your solution.
- Your application should run correctly against the sample data provided in the requirements document.
- Submit your code following the instructions found under ‘Submission Rules’.

Tips for Success:

- **You may use AI tools** to assist in your development process, but you are expected to fully understand any solutions provided and make conscious design decisions. During the pair programming session, you will be asked to build upon your solution and explain your choices.
- **Include unit and/or end-to-end tests. We recommend using Test-Driven Development (TDD) where possible.**
- Keep your functions and classes small and focused.
- Demonstrate clean code practices, strong object-oriented skills, and understanding of SOLID principles.
- Submit production-ready code that is clean, well-documented, and easy to understand.

Submission Rules:

- Submit your code as a **single .zip file containing only source code, supporting files, and test files** - no compiled code.
 - o **Please push your code to a private repository on GitHub**, share the repository link with us, we will provide you with a GitHub account to share the repository with. Make sure to include all necessary files to build and run your application.
- Include a brief explanation of your design and assumptions, as well as detailed instructions on how to run your application and the required environment (e.g., Windows, Linux).
- **Do not** use any external libraries that directly solve the problem.
- **You may** use libraries or tools for building or testing (e.g., JUnit, NUnit, Pytest, Jest, etc.).
- **Do not** include any executable files in your submission. Our system blocks these for security reasons, which may delay your application. (E.g., asp, bat, class, cmd, com, cpl, dll, exe, jar, jsp, jse, pif, wscr, shs, sh, vb, vbe, vbs, ws, wsc, wsf, wsh & msi)
- If you were tasked to attempt the assessment in C#, **kindly avoid submitting your code as a .msi file**

What We Assess:

We will review your **code design, tests, and overall readability, among other criteria**. While the problems are relatively small, we expect production-quality code - something you would be comfortable running, maintaining, and evolving in a real-world environment. There's no need to over-engineer your solution, but we are looking for more than just a basic algorithm.

Integrity and Fairness:

At GIC, we value integrity and fairness. Please do not share or publish these problems, so everyone has an equal opportunity.

Timeline:

You have **7 calendar days** from the date you received these instructions to complete and submit your code. You are free to attempt and complete the assessment at any time within this 7-day period, and you do not need to finish it in one sitting - feel free to work on it in multiple sessions as your schedule allows. If you have questions about the assessment, please respond to the email from which the assessment link was sent, and we will do our best to assist you.

We wish you the best of luck with your assessment!